

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: DEPARTMENT OF CIVIL ENGINEERING

2. Subject Name: PAVEMENT ENGINEERING L-T-P: 3-0-0 Credits:3

3. Pre-requisites: NIL

4. Syllabus and reference books:

PAVEMENT MATERIALS: Soil - as embankment, subgrade and shoulder materials, classification; Physical properties; tests; specifications; **Aggregates** - used in granular, bituminous, cement treated and cement concrete layers; tests for different physical properties; specifications; **Bituminous binders** - Bitumen, emulsion, cutback, modified binders and foam bitumen: production; tests for physical properties; grading of bitumen, selection of grade of bitumen, specifications- IS/ASTM; **Bituminous mixes:** Types of mixes; Marshall mix design; Mix specifications; Introduction to Superpave mix design; **Cementitious materials:** Cement treated base; cement treated sub-base; stabilised soil, characterization; **PAVEMENT ANALYSIS AND DESIGN:** Design philosophy of flexible and rigid pavements; Design factors: Traffic factors - lane distribution factor, axle load spectrum, standard axle, equivalent axle load factor, vehicle damage factor, design life; material factors; environmental factors; performance models and criteria; **Flexible pavement design approaches:** Analysis of Flexible Pavements; Empirical and analytical design methods; AASHTO-1993, IRC:37; **Rigid pavement design:** Stresses in rigid pavements - Westergaard's theory and assumptions, stresses due to curling, stresses due to loading, frictional stresses; types and design of joints; Introduction to IRC:58 pavement design approach; **PAVEMENT CONSTRUCTION:** Construction overview; equipment used for highway construction; Quality Control aspects; **PAVEMENT MAINTENANCE AND REHABILITATION:** Structural and functional evaluation; condition indicators; maintenance and rehabilitation alternatives

1) TEXT BOOKS:

1. Kadiyali, L.R and Lal, N.B. Principles and Practices of Highway Engineering, Seventh Edition, Khanna Publishers, New Delhi, 2017.
2. Venkatramayya, C. Transportation Engineering Volume-I&II, Universities Press, 2016
3. Harold N Atkins, Highway Materials, Soils and Concretes, Pearson Ed, 2002
4. Huang, Y.H. Pavement Analysis and Design, Second Edition, Pearson, 2004.
5. Horonjeff, R., McKelvey, F. X., Sproule, W. J., and Young, S. B. Planning and Design of Airports, Fifth Edition, McGraw-Hill, New York, 2010.

REFERENCES:

1. IRC:37 – 2018: Guidelines for the Design of Flexible Pavements
2. IRC:58 – 2015: Guidelines for the Design of Plain Jointed Rigid Pavements for Highways
3. Asphalt Institute Manual Series – 2, Asphalt Mix Design Methods
4. Other relevant Indian Roads Congress codes

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to Pavement Engineering: Pavement types; pavement composition; function of different pavement layers; Functional and structural performance of different types of pavements	03
2.	PAVEMENT MATERIALS: Soil - as embankment, subgrade and shoulder materials, classification; Physical properties; tests; specifications Aggregates - used in granular, bituminous, cement treated and cement concrete layers; tests for different physical properties; specifications Bituminous binders - Bitumen, emulsion, cutback, modified binders and foam bitumen: production; tests for physical properties; grading of bitumen, selection of grade of bitumen, specifications- IS/ASTM Bituminous mixes: Types of mixes; Marshall mix design; Mix specifications; Introduction to Superpave mix design	15

	Cementitious materials: Cement treated base; cement treated sub-base; stabilised soil, characterization	
3.	PAVEMENT DESIGN: Design philosophy of flexible and rigid pavements Design factors: Traffic factors - lane distribution factor, axle load spectrum, standard axle, equivalent axle load factor, vehicle damage factor, design life; material factors; environmental factors; performance models and criteria. Flexible pavement design approaches: Analysis of Flexible Pavements; Empirical and analytical design methods; AASHTO-1993, IRC:37 Rigid pavement design: Stresses in rigid pavements - Westergaard's theory and assumptions, stresses due to curling, stresses due to loading, frictional stresses; types and design of joints; Introduction to IRC:58 pavement design approach	12
4.	PAVEMENT CONSTRUCTION: Construction overview; equipment used for highway construction; Quality Control aspects	03
5.	PAVEMENT MAINTENANCE AND REHABILITATION Structural and functional evaluation; condition indicators; maintenance and rehabilitation alternatives.	03
Total number of hours		36